

Model Performance Comparison (Test Set)
Values shown as Mean (95% CI)

Model	AUC	AUPRC	Accuracy	Sensitivity	Specificity	F1	Ppv	Npv	Sens At 90 Spec	Sens At 70 Spec
Inception3D	0.546 (0.479-0.613)	0.420 (0.351-0.490)	0.600 (0.544-0.656)	0.142 (-0.252-0.537)	0.881 (0.549-1.212)	0.106 (-0.188-0.400)	0.084 (-0.150-0.319)	0.635 (0.594-0.676)	0.105 (0.040-0.171)	0.337 (0.203-0.470)
Levit-RI	0.476 (0.450-0.502)	0.369 (0.334-0.404)	0.382 (0.376-0.388)	1.000 (nan-nan)	0.003 (-0.006-0.012)	0.552 (0.549-0.554)	0.381 (0.379-0.383)	0.200 (-0.355-0.755)	0.058 (-0.013-0.129)	0.253 (0.193-0.312)
Levit-Attention	0.514 (0.496-0.532)	0.395 (0.358-0.433)	0.476 (0.339-0.613)	0.600 (0.057-1.143)	0.400 (-0.153-0.953)	0.386 (0.098-0.675)	0.307 (0.093-0.522)	0.498 (0.150-0.846)	0.079 (0.006-0.152)	0.295 (0.213-0.376)
3Dcnn	0.526 (0.500-0.553)	0.384 (0.370-0.399)	0.524 (0.361-0.687)	0.400 (-0.280-1.080)	0.600 (-0.080-1.280)	0.220 (-0.154-0.595)	0.152 (-0.106-0.410)	0.372 (-0.050-0.794)	0.079 (0.046-0.112)	0.274 (0.192-0.356)
Cnnlstm	0.501 (0.466-0.535)	0.374 (0.362-0.387)	0.524 (0.361-0.687)	0.400 (-0.280-1.080)	0.600 (-0.080-1.280)	0.220 (-0.154-0.595)	0.152 (-0.106-0.410)	0.372 (-0.050-0.794)	0.089 (0.072-0.107)	0.242 (0.188-0.296)
Attention Pool Noinitweight	0.832 (0.806-0.858)	0.764 (0.704-0.823)	0.770 (0.680-0.860)	0.500 (0.177-0.823)	0.935 (0.881-0.990)	0.587 (0.317-0.857)	0.844 (0.785-0.903)	0.764 (0.657-0.872)	0.732 (0.625-0.838)	0.800 (0.782-0.818)
Attention Pool Extra3	0.885 (0.870-0.900)	0.868 (0.843-0.892)	0.832 (0.767-0.897)	0.705 (0.445-0.965)	0.910 (0.837-0.983)	0.745 (0.575-0.916)	0.845 (0.759-0.930)	0.845 (0.747-0.943)	0.774 (0.702-0.845)	0.874 (0.831-0.916)
Mean Pool Extra3	0.878 (0.847-0.908)	0.866 (0.828-0.905)	0.800 (0.754-0.846)	0.700 (0.494-0.906)	0.861 (0.712-1.010)	0.721 (0.645-0.797)	0.787 (0.653-0.921)	0.835 (0.751-0.918)	0.726 (0.629-0.823)	0.868 (0.828-0.908)
Singletask	0.822 (0.773-0.871)	0.750 (0.680-0.820)	0.770 (0.704-0.836)	0.495 (0.305-0.684)	0.939 (0.898-0.980)	0.609 (0.443-0.775)	0.834 (0.755-0.913)	0.755 (0.689-0.821)	0.595 (0.439-0.751)	0.816 (0.739-0.892)
Uniform Extra3	0.869 (0.854-0.884)	0.846 (0.822-0.871)	0.800 (0.785-0.815)	0.700 (0.526-0.874)	0.861 (0.772-0.950)	0.721 (0.653-0.789)	0.770 (0.688-0.853)	0.831 (0.763-0.899)	0.726 (0.672-0.781)	0.853 (0.835-0.871)
No RI Full Train	0.830 (0.798-0.863)	0.764 (0.689-0.839)	0.736 (0.638-0.834)	0.489 (0.115-0.864)	0.887 (0.749-1.025)	0.527 (0.148-0.906)	0.602 (0.165-1.039)	0.754 (0.648-0.860)	0.642 (0.551-0.733)	0.800 (0.732-0.868)
Original Noinitweights	0.837 (0.811-0.862)	0.773 (0.720-0.827)	0.740 (0.604-0.876)	0.611 (0.170-1.051)	0.819 (0.546-1.093)	0.579 (0.173-0.986)	0.577 (0.146-1.008)	0.804 (0.670-0.938)	0.684 (0.615-0.754)	0.821 (0.758-0.884)
Original	0.842 (0.827-0.858)	0.756 (0.691-0.820)	0.744 (0.652-0.836)	0.679 (0.320-1.038)	0.784 (0.557-1.011)	0.638 (0.401-0.875)	0.711 (0.555-0.866)	0.825 (0.704-0.946)	0.674 (0.559-0.788)	0.837 (0.801-0.873)
Original Test	0.838 (0.815-0.862)	0.770 (0.729-0.810)	0.690 (0.508-0.872)	0.611 (0.215-1.006)	0.739 (0.319-1.159)	0.571 (0.290-0.852)	0.682 (0.441-0.922)	0.775 (0.670-0.881)	0.674 (0.606-0.741)	0.842 (0.819-0.865)
Vivit	0.498 (0.434-0.561)	0.374 (0.332-0.415)	0.556 (0.426-0.686)	0.211 (-0.338-0.759)	0.768 (0.228-1.308)	0.126 (-0.172-0.424)	0.109 (-0.099-0.317)	0.490 (0.150-0.831)	0.063 (0.026-0.100)	0.253 (0.121-0.385)